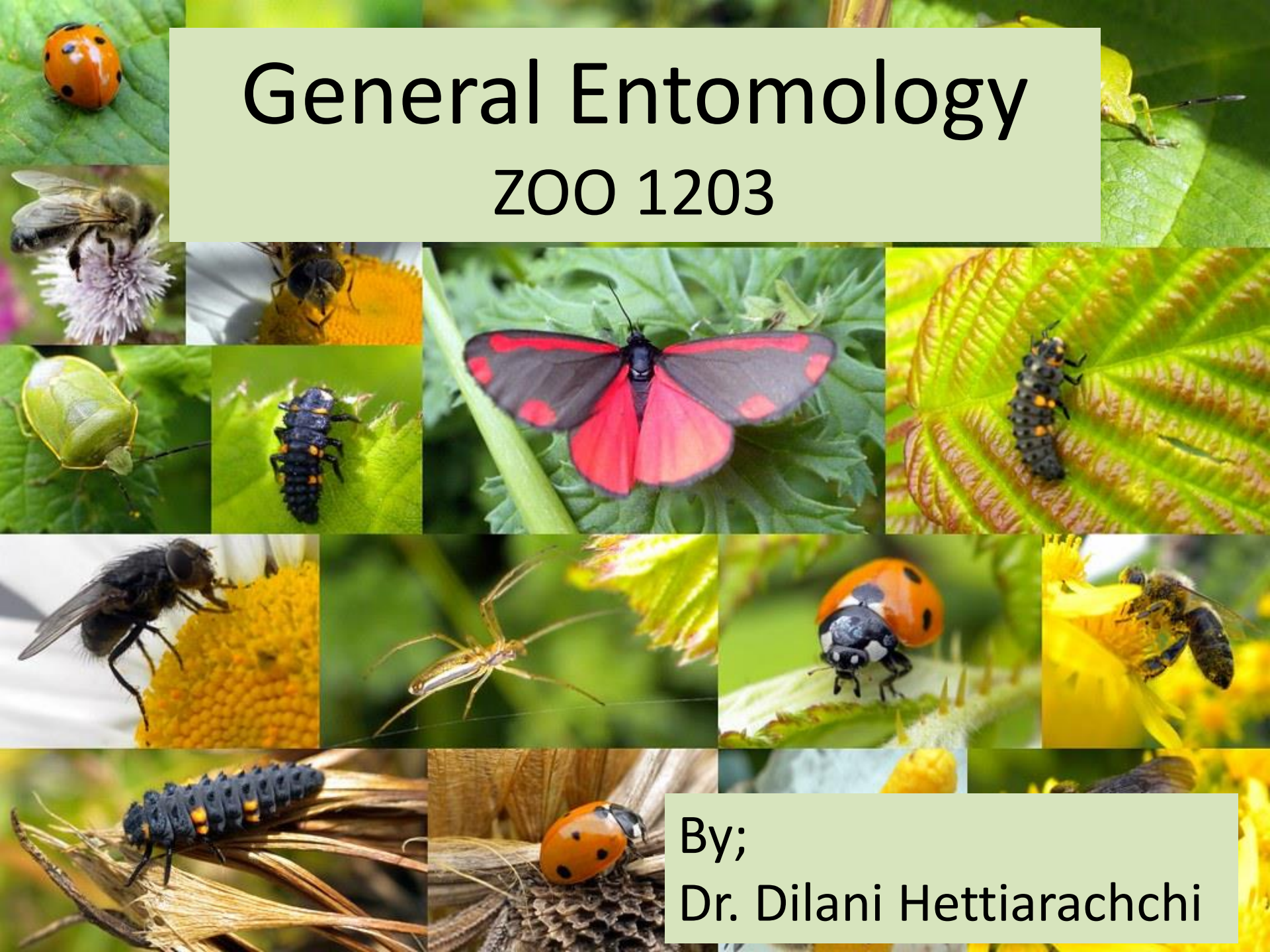


General Entomology

ZOO 1203



By;
Dr. Dilani Hettiarachchi

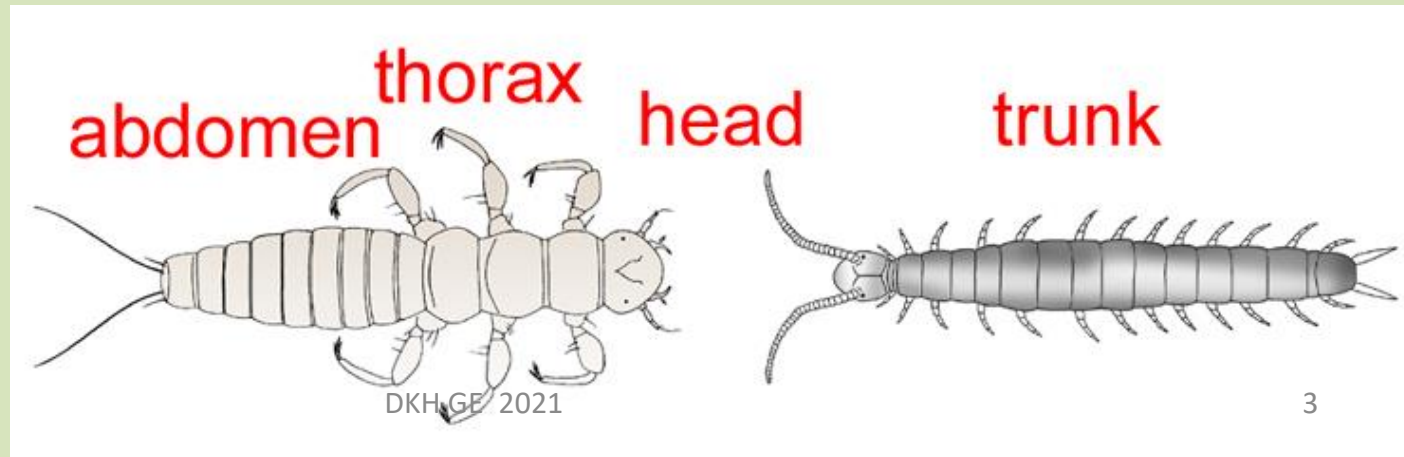
Characteristics of Phylum Arthropoda

- Arthropoda (Gr. *arthron*, joint, *podos*, foot)
- Free-living, parasitic, marine, freshwater, terrestrial
- Size - Adults range from 0.1 mm to 4 m
- Most extensive phylum in the animal kingdom, composed of $>3/4$ of all known animals approx. 900,000
- Origin- oldest fossils around 550 my old



General features

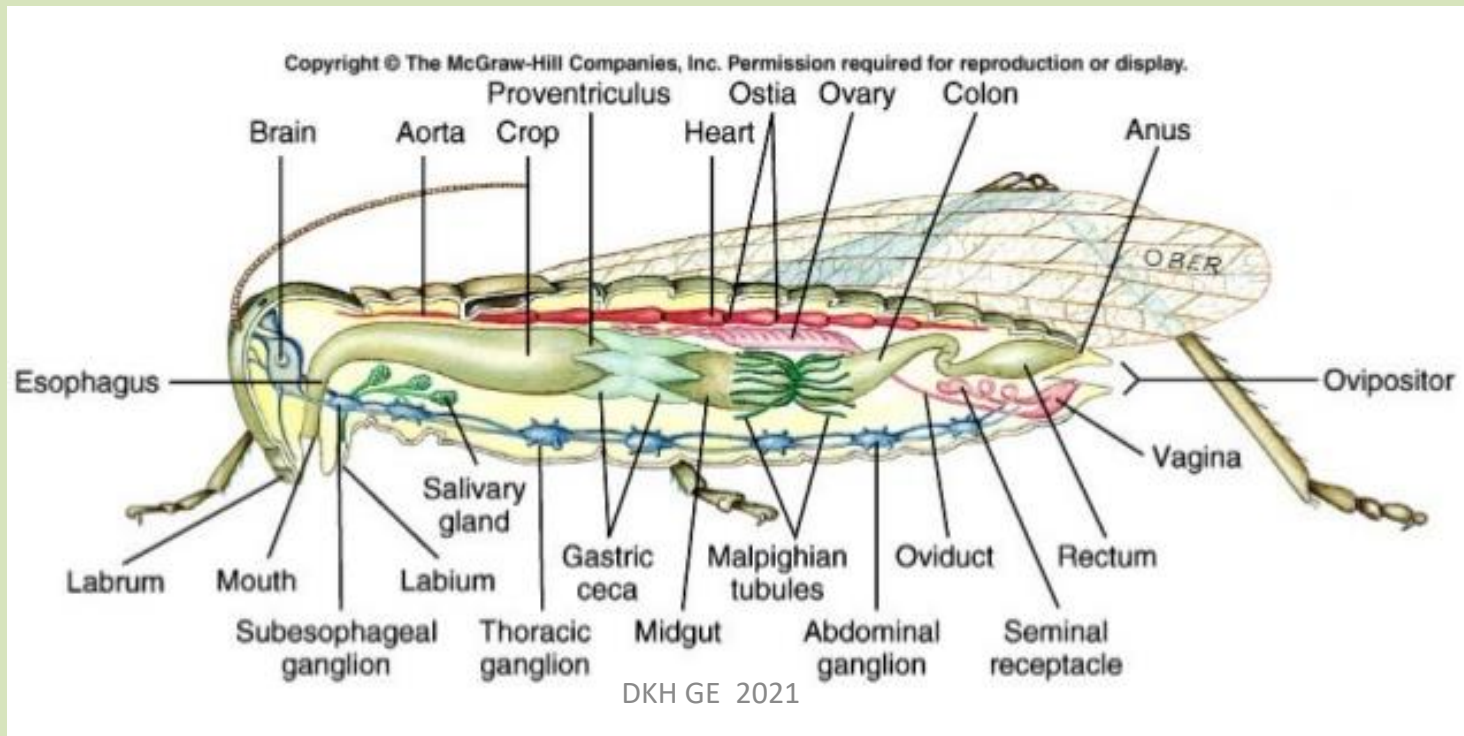
- Bilateral symmetry
- **Metamerically segmented** body divided into **tagmata** - head and trunk or head, thorax and abdomen
- Organ system grade of body organization
- **Jointed appendages** - modified for specialized functions



- Body covered with an **exoskeleton** i.e. cuticle
- **Exoskeleton** contains chitin in combination with protein, lipid, and often calcium carbonate secreted by underlying epidermis
- **Exoskeleton is shed** (molt/ecdysis) at intervals
- Bears a body cavity – coelom
- **Coelom reduced** in adults
- Most of the body cavity consists of **hemocoel** (cavities, or spaces, in the tissues) filled with blood



- All arthropods have;
 - Digestive system
 - Open circulatory system
 - Ventral nerve system
 - Breathe by gills, trachea or spiracles



Class Insecta

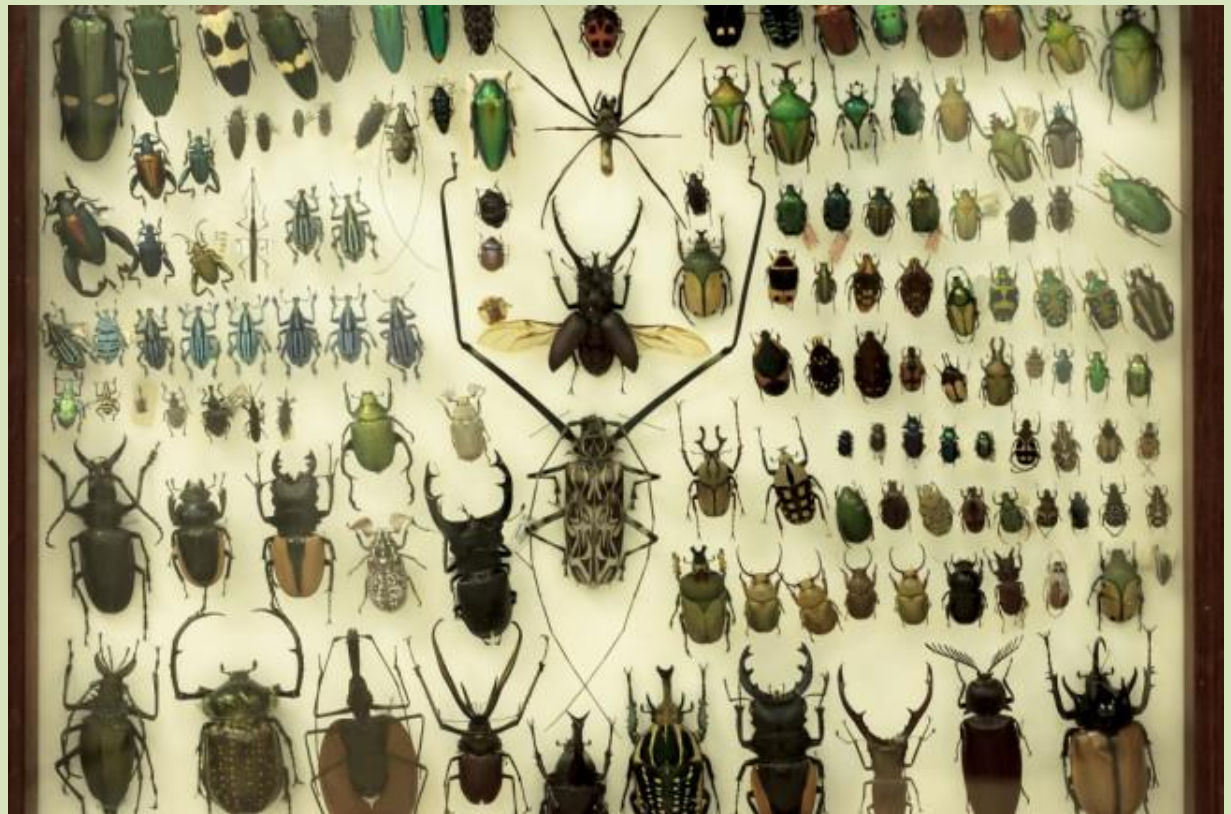
- Shows tagmata – body divided into three parts; head, thorax and abdomen
- One pair of antenna
- A pair of mandibles
- A pair of maxillae
- A pair of compound eyes

- Thorax consists of three segments
- Each segment have a pair of legs
- Second and third thoracic segments generally bear two pairs of wings



Quiz:

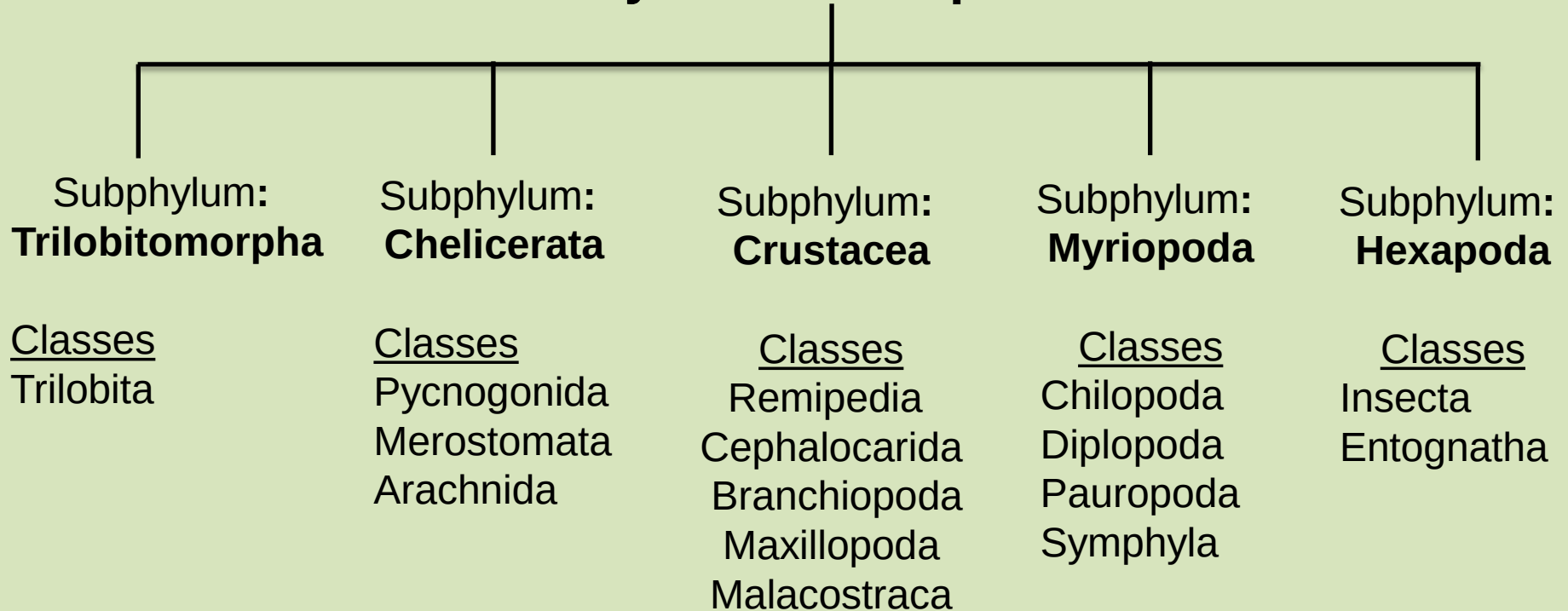
- Name the classes of the Phylum Arthropoda and write down the salient features of each of the classes



Phylum: Arthropoda

Classification

Phylum: Arthropoda



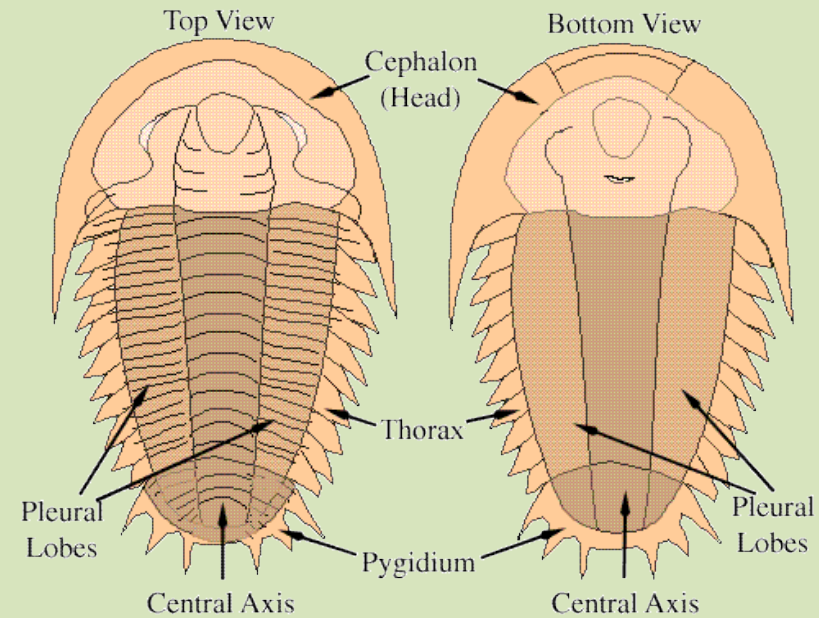
Classification

SUBPHYLUM: TRILOBITOMORPHA

- Lived in the Cambrian (520 mya) and Permian (250 mya) periods
- Trilobe shape body
- Bottom dwellers and probably scavengers
- Size - 2 to 67 cm in length
- Exoskeleton contained chitin, strengthened by calcium carbonate
- Three tagma: head (cephalon), thorax, and pygidium
- Thorax - variable number of somites (segments)
- Somites of the pygidium - fused into a plate

SUBPHYLUM: TRILOBITOMORPHA

- Head bears a pair of antennae, compound eyes, mouth, and four pairs of jointed appendages
- Each body somite except the last also bears a pair of biramous (two-branched) appendages
- One of the branches - fringe of filaments that may be gills
- Single class- Trilobita
e.g. Trilobite



SUBPHYLUM: CHELICERATA

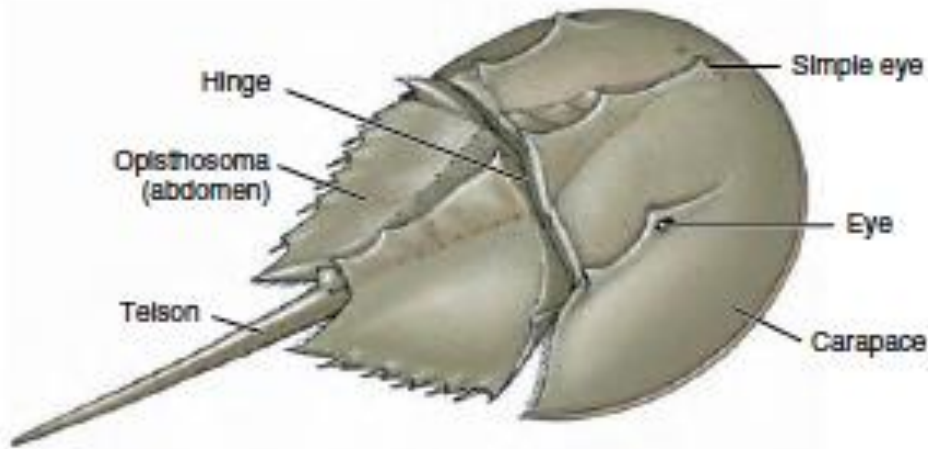
- Ancient group that includes extinct and extant groups (e.g. horseshoe crabs, spiders, ticks and mites, scorpions, sea spiders)
- Six pairs of appendages - pair of chelicerae, a pair of pedipalps, and four pairs of walking legs
- No mandibles and no antennae
- Suck liquid food from their prey
- Three classes: Merostomata, Pycnogonida, Arachnida,

SUBPHYLUM: CHELICERATA

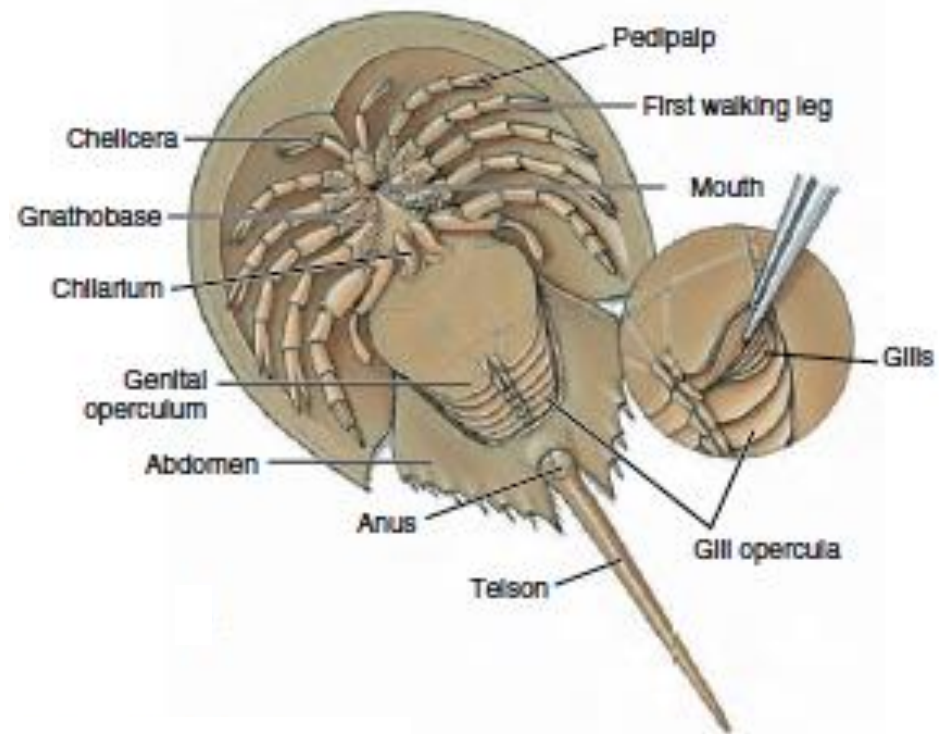
CLASS: MEROSTOMATA

- Most are extinct, only the Horseshoe crabs are living
- An ancient marine group, dates from the Cambrian period – living fossils
- All are free living and marine species
- Body two regions – Cephalothorax, abdomen
- **Cephalothorax** (head & thorax fused) bears **five pairs** of walking legs and a pair of chelicerae
- An unsegmented horseshoe-shaped **carapace** (upper part of shell) covers the **cephalothorax** and abdomen
- Abdomen bears - a long **telson**/tailpiece

SUBPHYLUM: CHELICERATA, CLASS: MEROSTOMATA



Horseshoe crab (*Limulus* sp.)



SUBPHYLUM: CHELICERATA, CLASS: MEROSTOMATA

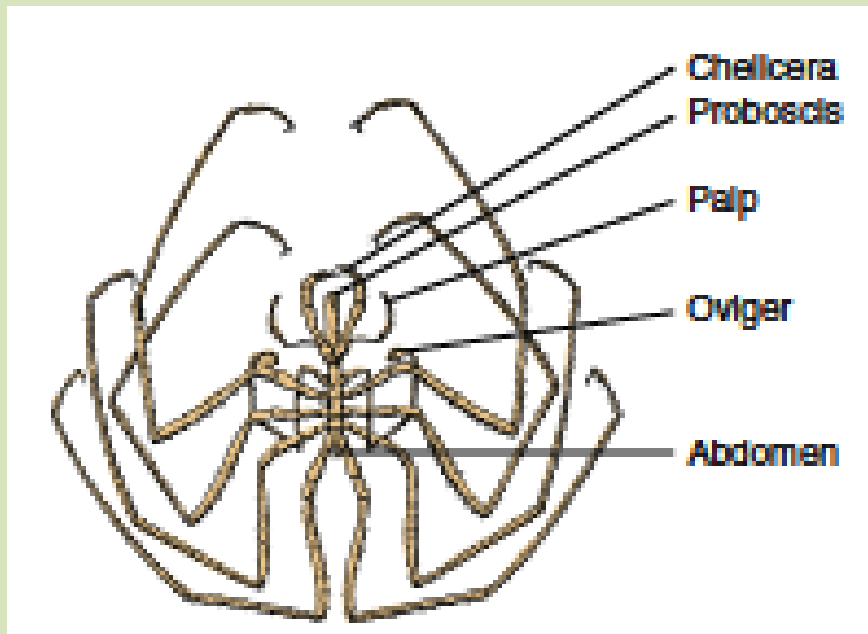
- Some of the abdominal appendages bears **book gills** (flat, leaf-like gills)
- Two compound and two simple eyes present on the carapace
- Swims - abdominal plates
- Walk with walking legs
- Reproduction: sexual, external fertilization
- Larval stage present – trilobite larva

SUBPHYLUM: CHELICERATA, CLASS: PYCNOGONIDA

- All marine
- Size – a few millimeters to few centimeters
- Have small, thin bodies
- Four pairs of long, thin walking legs
- Bears chelicerae and pedipalps
- Mouth is at the tip of a long **proboscis** - sucks juices from cnidarians and soft-bodied animals
- Have four simple eyes
- Circulatory system limited to a simple dorsal heart
- Excretory and respiratory systems are absent

SUBPHYLUM: CHELICERATA, CLASS: PYCNOGONIDA

- Diffusion of gases and wastes takes place through the long, thin body and legs
- The digestive system sends branches into the legs
- Commonly known as sea spiders



SUBPHYLUM: CHELICERATA, CLASS: ARACHNIDA

- Includes scorpions, pseudoscorpions, whip scorpions, ticks, mites, harvestmen
- All are terrestrial, free living or parasitic
- Body tagma – cephalothorax and abdomen
- Cephalothorax - bears a pair of chelicerae, a pair of pedipalps, and four pairs of walking legs
- Lack antennae and mandibles
- Most are predaceous and have pedipalps and chelicerae with poison glands, or stingers

SUBPHYLUM: CHELICERATA, CLASS: ARACHNIDA

- Usually have sucking mouth parts, ingest the fluids and soft tissues from the bodies of their prey
- More than 70,000 species
- First of the arthropods to move into terrestrial habitats
- Reproduction – sexual, sexes separate, internal fertilization
- Many orders of Arachnids
 - Order Araneae: Spiders
 - Order Opiliones: Harvestmen
 - Order Scorpionida: Scorpions
 - Order Acari: Ticks and Mites

SUBPHYLUM: CHELICERATA, CLASS: ARACHNIDA



Order Araneae: Spider



Order Opiliones: Harvestmen



Order Scorpionida: Scorpion



Order Acari: Tick

SUBPHYLUM: CHELICERATA, CLASS: ARACHNIDA



Order Thelyphonida: Whip Scorpion



Order Amblypygi: Whip Spiders



Order Pseudoscorpiones: Pseudoscorpion



Order Acari: Mite

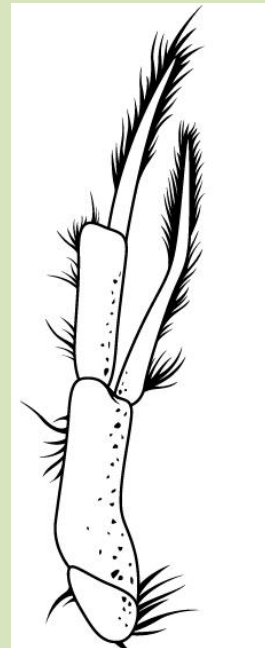
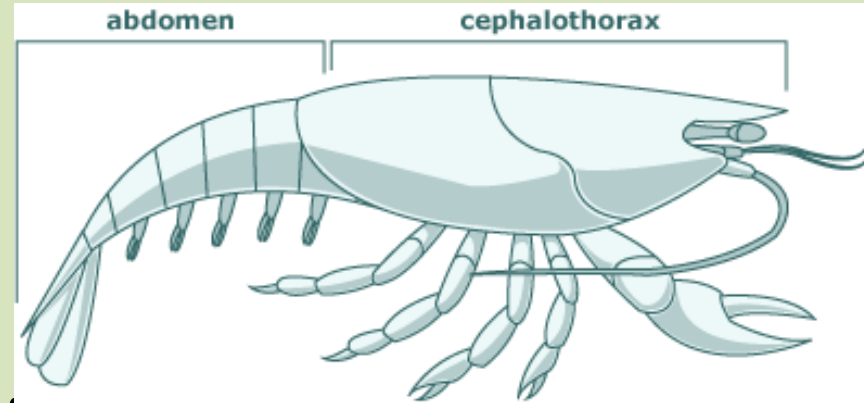
SUBPHYLUM: CRUSTACEA

- Latin *crusta* = shell
- Bears a hard external shell
- Over 30,000 species
- Mostly free living, mainly aquatic marine and fresh water, few terrestrial
- Only arthropods with **two pairs of antennae**
- Have a pair of mandibles and two pairs of maxillae on head
- **Mandibles** and **maxillae** – involved in sensory, mastication, and food-handling



SUBPHYLUM: CRUSTACEA

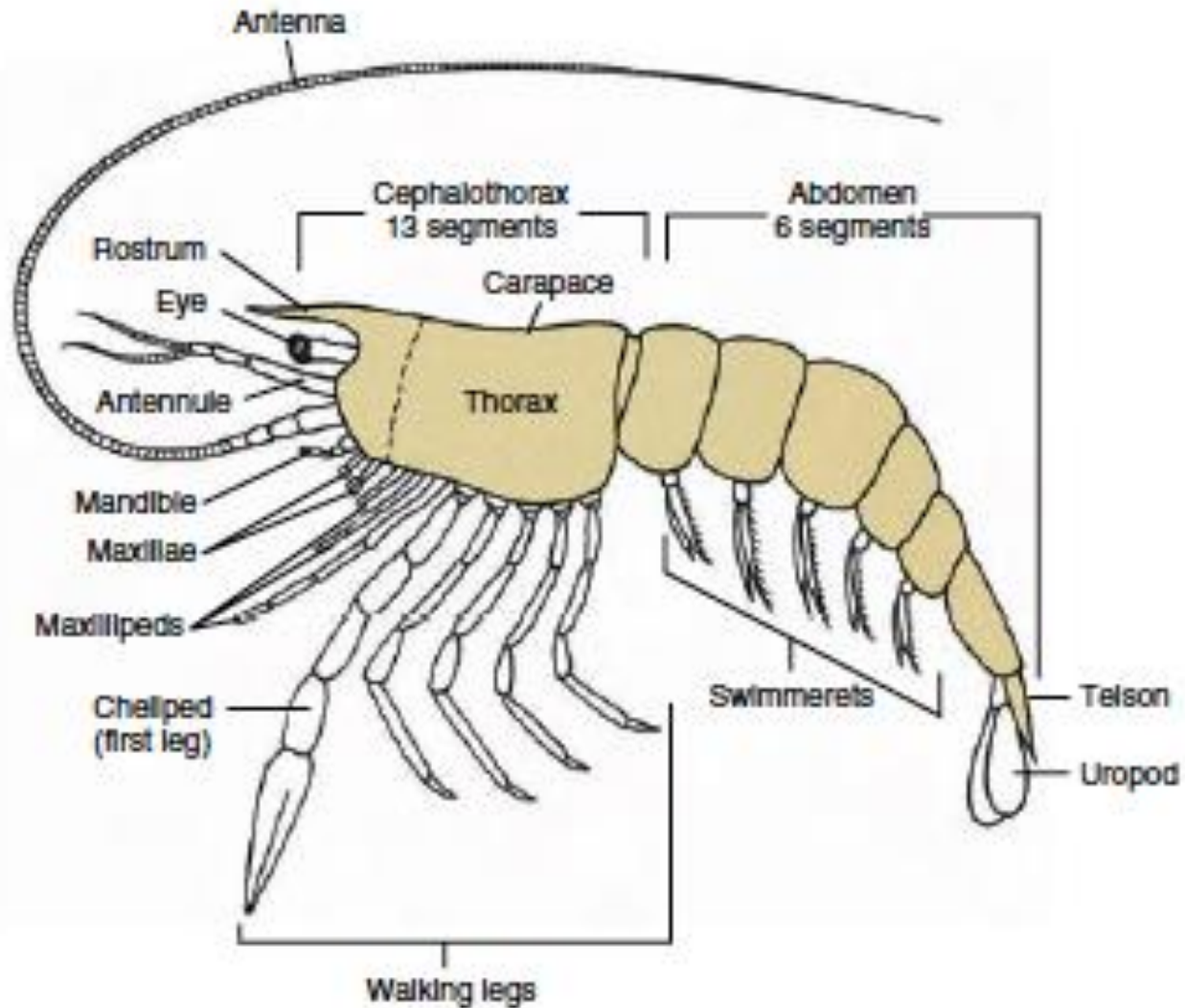
- Some - head, thorax, abdomen
- In most - thoracic segments are fused with the head to form a **cephalothorax**
- Anterior end – non-segmented **rostrum**
- posterior end – non-segmented **telson**
- **Uropods** forms a tail fin in many forms
- Thoracic and abdominal appendages used for walking or swimming
- All appendages are **biramous** (two main branches)



SUBPHYLUM: CRUSTACEA

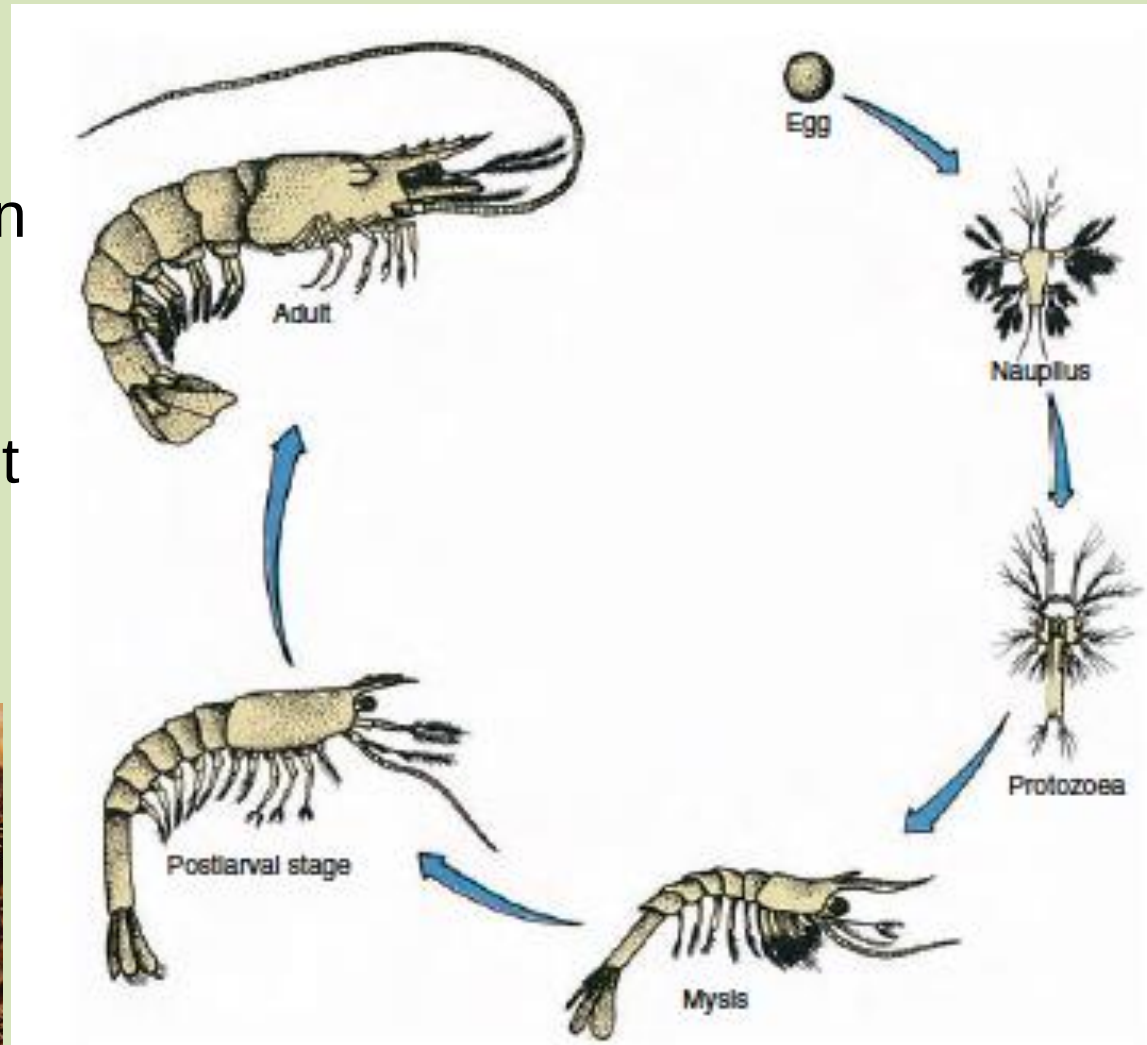
- Dorsal cuticle of the head extend posteriorly and around the sides to cover or be fused with some or all of the thoracic and abdominal segments – **carapace/shell**
- Bodies covered with a cuticle is composed of chitin, protein and calcareous material
- Respiration – gills (delicate, featherlike projections with very thin cuticle)
- Sensory structures – compound eye, antennae
- Food & feeding – predatory, filter feeders, scavengers

SUBPHYLUM: CRUSTACEA



SUBPHYLUM: CRUSTACEA

- Reproduction
 - sexes separate
 - sexual reproduction
- Direct as well as indirect development with larval stages



SUBPHYLUM: CRUSTACEA

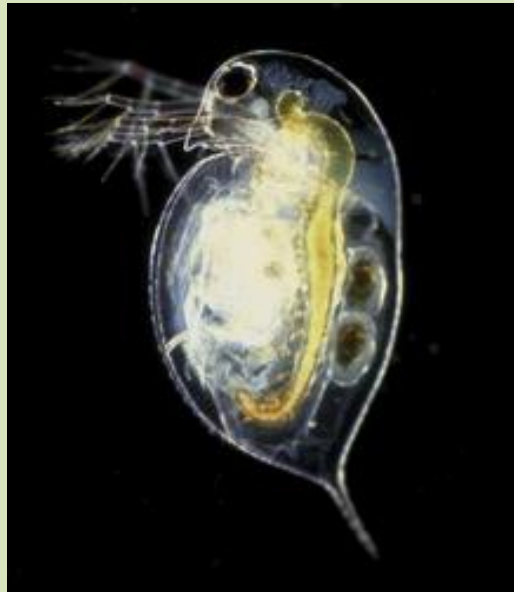
- **Classes**

- Remipedia: remipede
- Cephalocarida: cephalocarid crustacean
- Branchiopoda: Tadpole shrimp, water flea
- Maxillopoda: Ostracod, Copepods, barnacles
 - Malacostraca: isopods/woodlouse, crabs, shrimps, lobsters

SUBPHYLUM: CRUSTACEA



Tadpole shrimp



Water flea



Ostracod



Copepod



Barnacles

SUBPHYLUM: CRUSTACEA



Wood louse (isopod)



Lobster



Crab



Shrimp

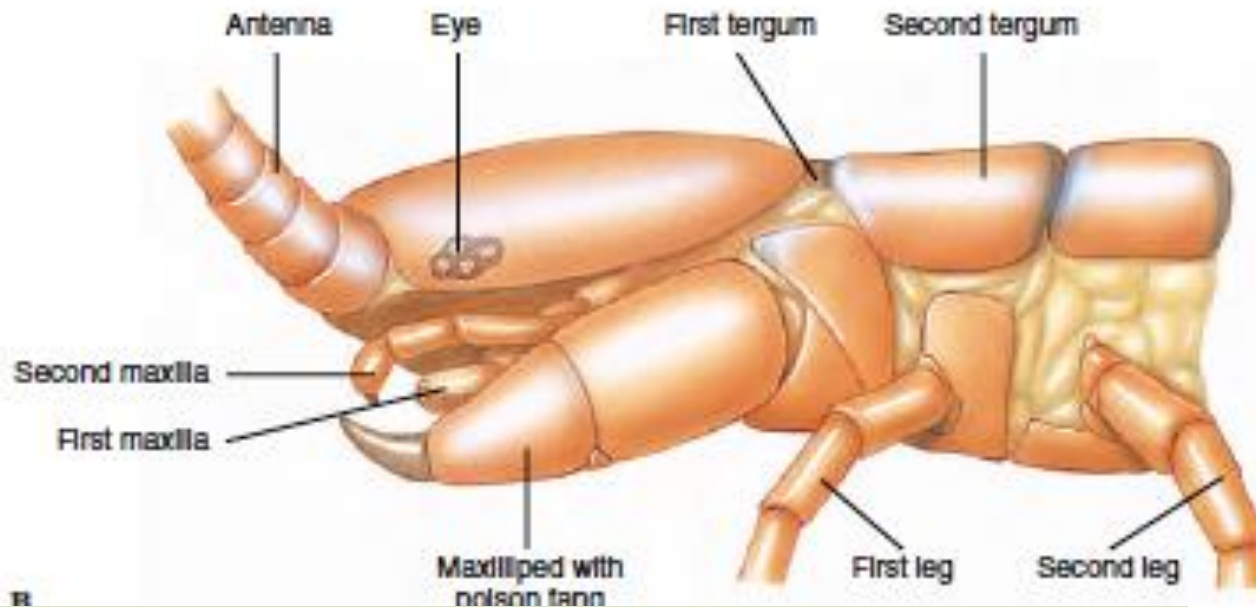
SUBPHYLUM: MYRIAPODA

- More than 3 pairs of legs (usually 10 -750 pairs)
- One pair of antennae
- Tracheae carry respiratory gases to body tissues
- Excretion is usually by malpighian tubule
- Eyes usually simple, few have compound eyes
- Classes - Chilopoda: centipedes
 - Diplopoda: millipedes
 - Pauropoda
 - Symphyla

UBPHYLUM: MYRIAPODA, CLASS: CHILOPODA

- Terrestrial forms
- Have flattened bodies with a few to ~200 segments
- Each segment, except the one behind the head and the last two in the body, bears a pair of jointed legs
- Appendages of the first body segment modified to form poison claws
- Head: a pair of antennae, a pair of mandibles, and one or two pairs of maxillae, pair of compound eyes on the dorsal sides
- Reproduction - Sexes are separate, fertilization internal, development direct

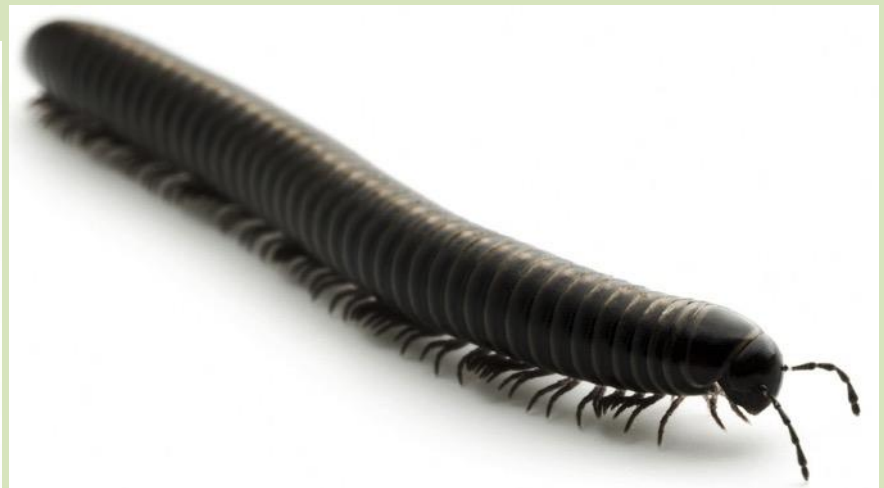
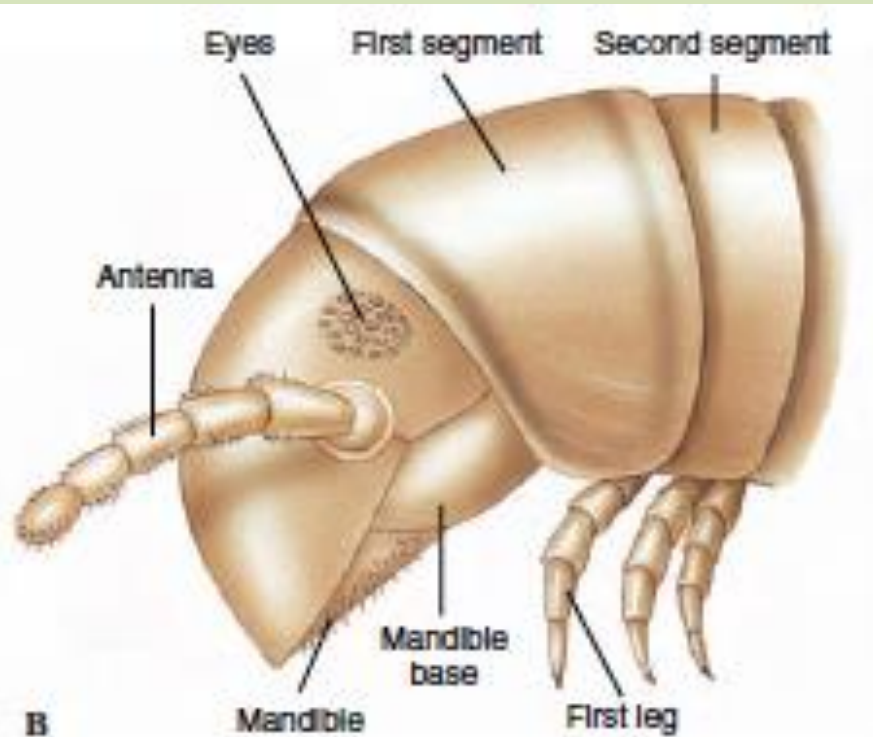
SUBPHYLUM: MYRIAPODA, CLASS: CHILOPODA
e.g. Centipede



SUBPHYLUM: MYRIAPODA, CLASS: DIPLOPODA

- Terrestrial forms
- More than 12000 known species
- Each abdominal segment bears two pairs of jointed legs
- Body - cylindrical, consists of 25 to 100 segments
- Thorax (first three segments behind head) short, consists of four somites, each bear one pair of legs
- Head bears two clumps of simple eyes and a pair each of antennae, mandibles, and maxillae
- Reproduction – sexes separate, sexual, fertilization internal, development indirect

SUBPHYLUM: MYRIAPODA, CLASS: DIPLOPODA



SUBPHYLUM: HEXAPODA

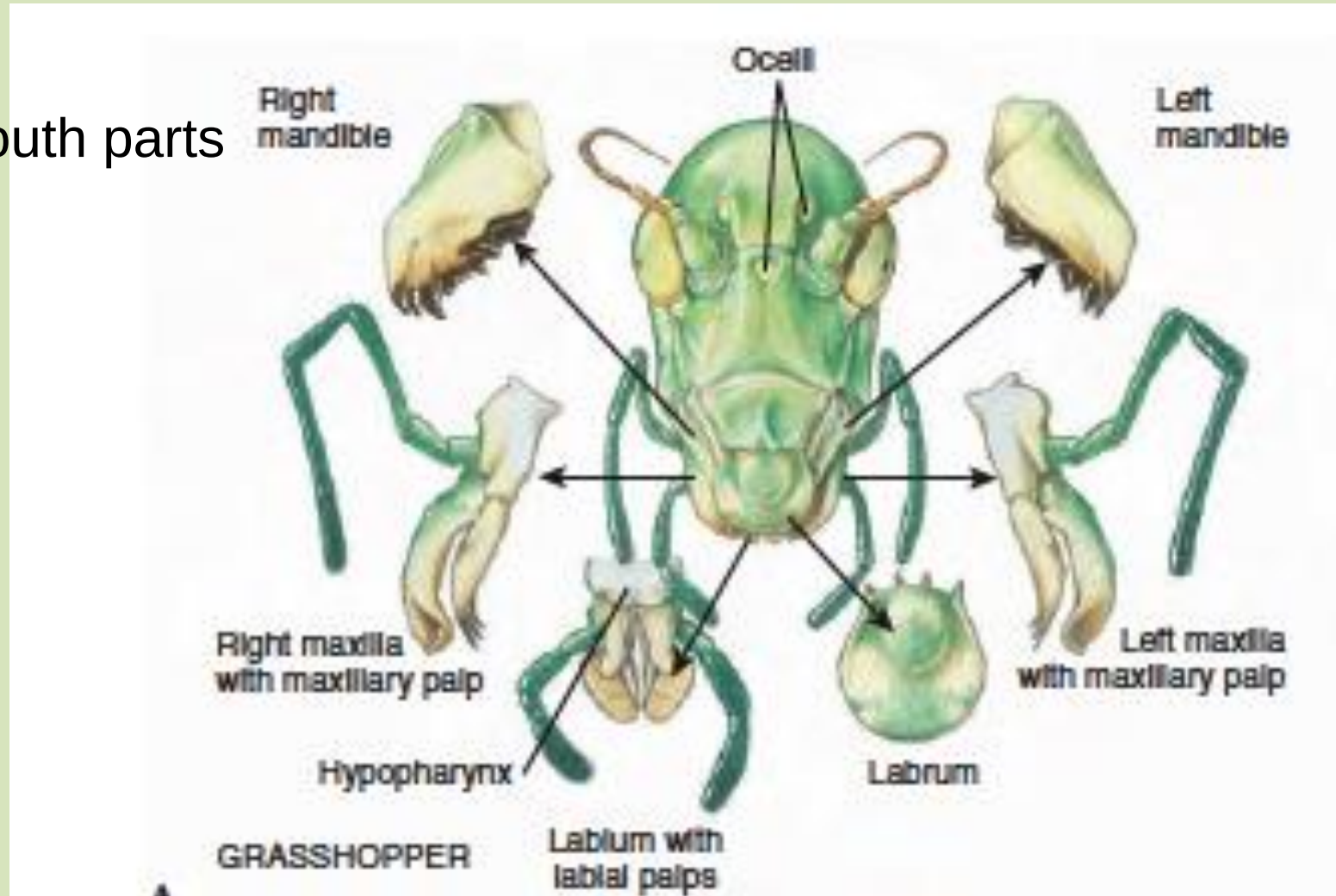
- Three pairs of legs
- Body divided into head, thorax and abdomen
- One pair of antennae
- Tracheae carry respiratory gases to body tissues
- Excretion is usually by malpighian tubule
- Eyes - compound eyes
- Classes - Insecta: Insects
 - Entognatha: e.g. Springtails

SUBPHYLUM: HEXAPODA, CLASS: INSECTA

- Most diverse and abundant of all groups of arthropods
- Number of estimated species -10 million
- Habitat – terrestrial, avian, aquatic, few marine
- Free living, parasitic
- Body tagmata - head, thorax, and abdomen
- Head bears a pair of relatively large compound eyes, a pair of antennae, and usually three ocelli
- Mouthparts - formed from hardened cuticle, consist of a labrum, a pair each of mandibles and maxillae, a labium, and a tongue-like hypopharynx

SUBPHYLUM: HEXAPODA, CLASS: INSECTA

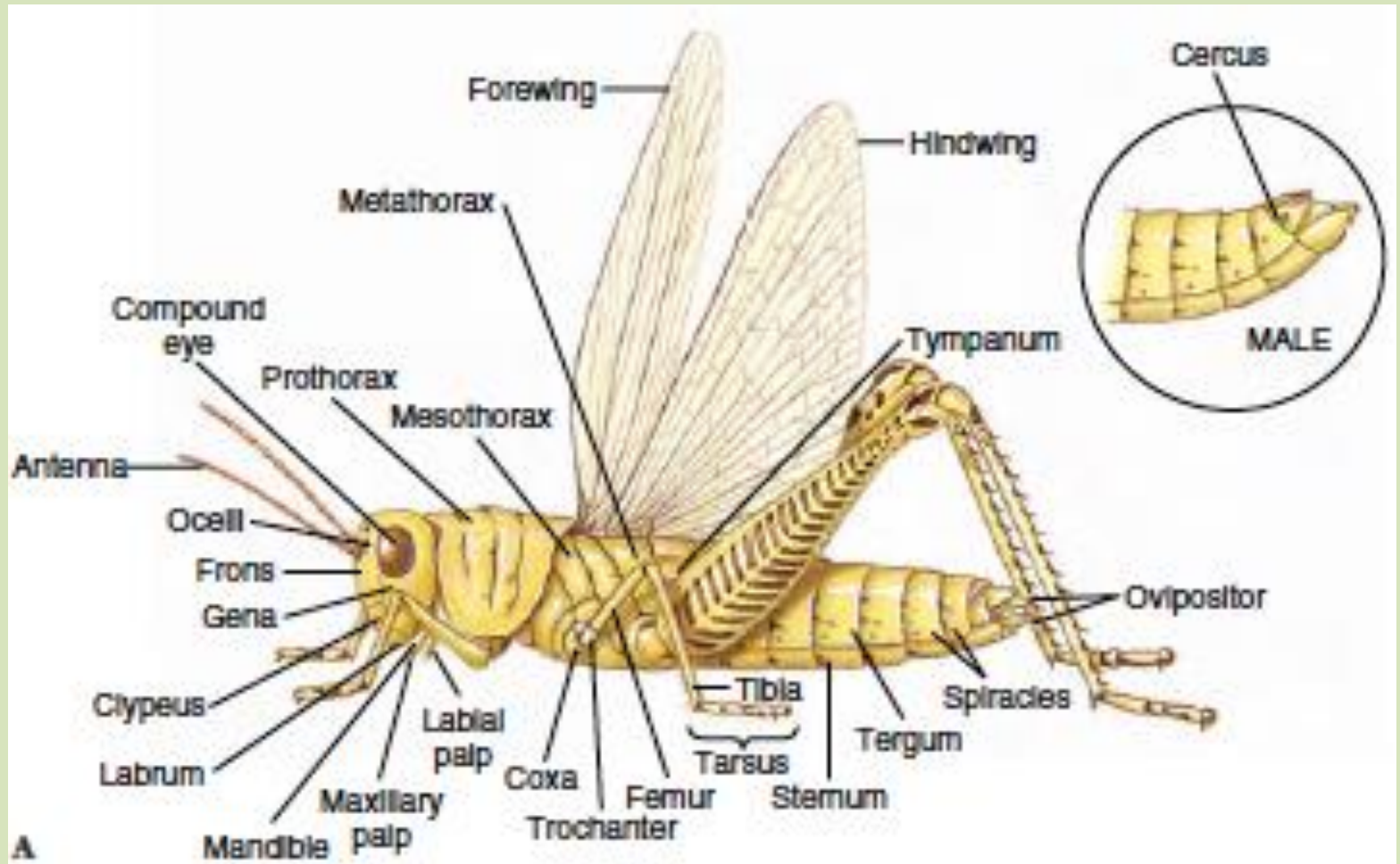
Mouth parts



SUBPHYLUM: HEXAPODA, CLASS: INSECTA

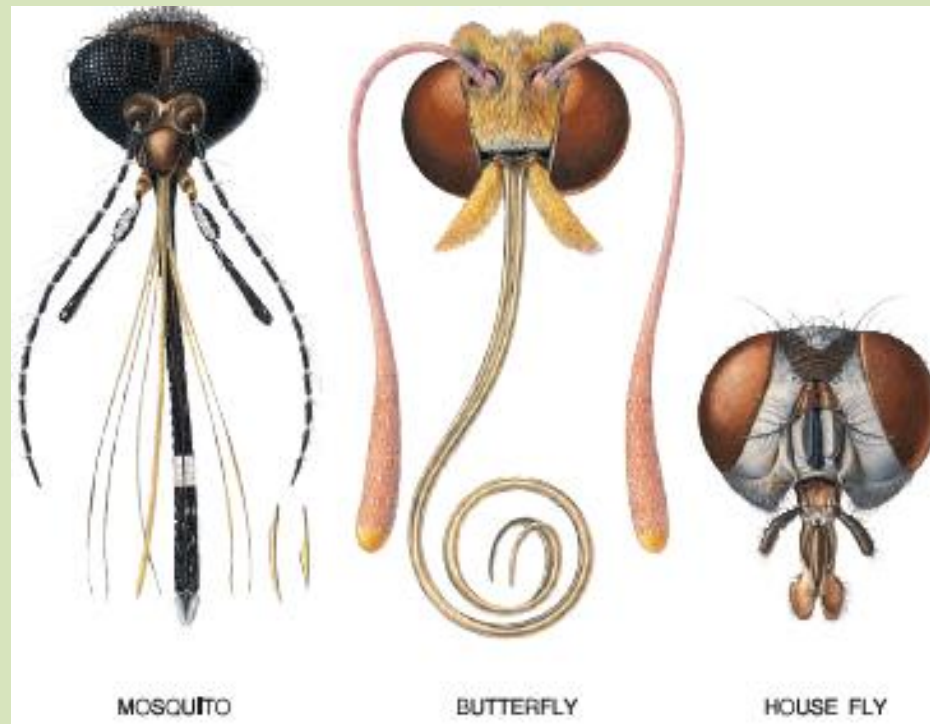
- Thorax is composed of three somites: prothorax, mesothorax, metathorax, each bears a pair of legs
- Most insects mesothorax and metathorax each bear a pair of wings
- Abdomen of insects is composed of 9 to 11 segments
- Locomotion – walking, flying, swimming
- Respiration – tracheal system

SUBPHYLUM: HEXAPODA, CLASS: INSECTA



SUBPHYLUM: HEXAPODA, CLASS: INSECTA

- Food & feeding- herbivorous, carnivorous, parasitic
- Mouth parts –adapted to the type of diet



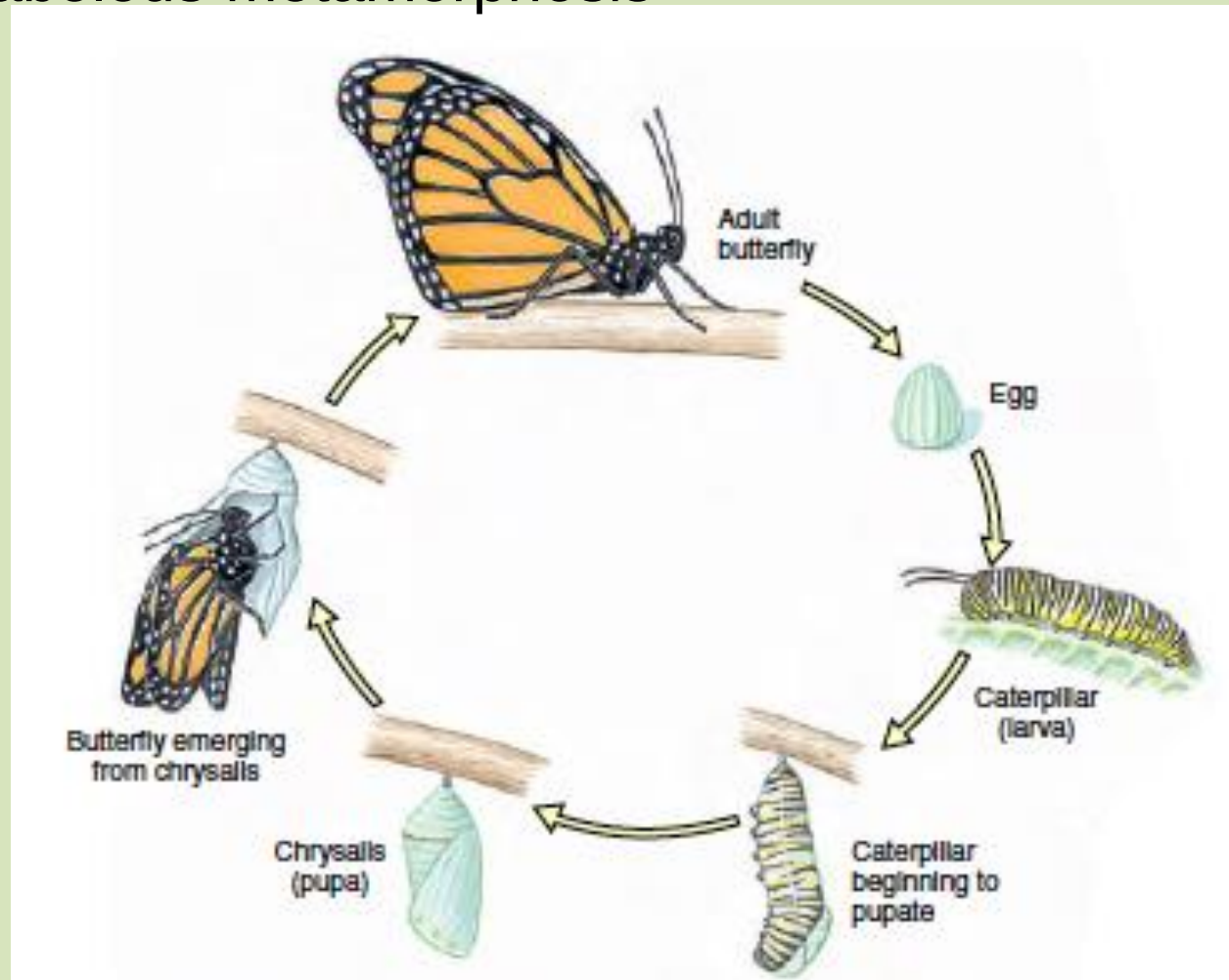
SUBPHYLUM: HEXAPODA, CLASS: INSECTA

- Sensory structures – compound eyes, ocelli, antennae
- Reproduction : sexes separate, fertilization internal, development indirect, larval stages present
- Larvae undergo metamorphosis – holometabolous , hemimetabolous



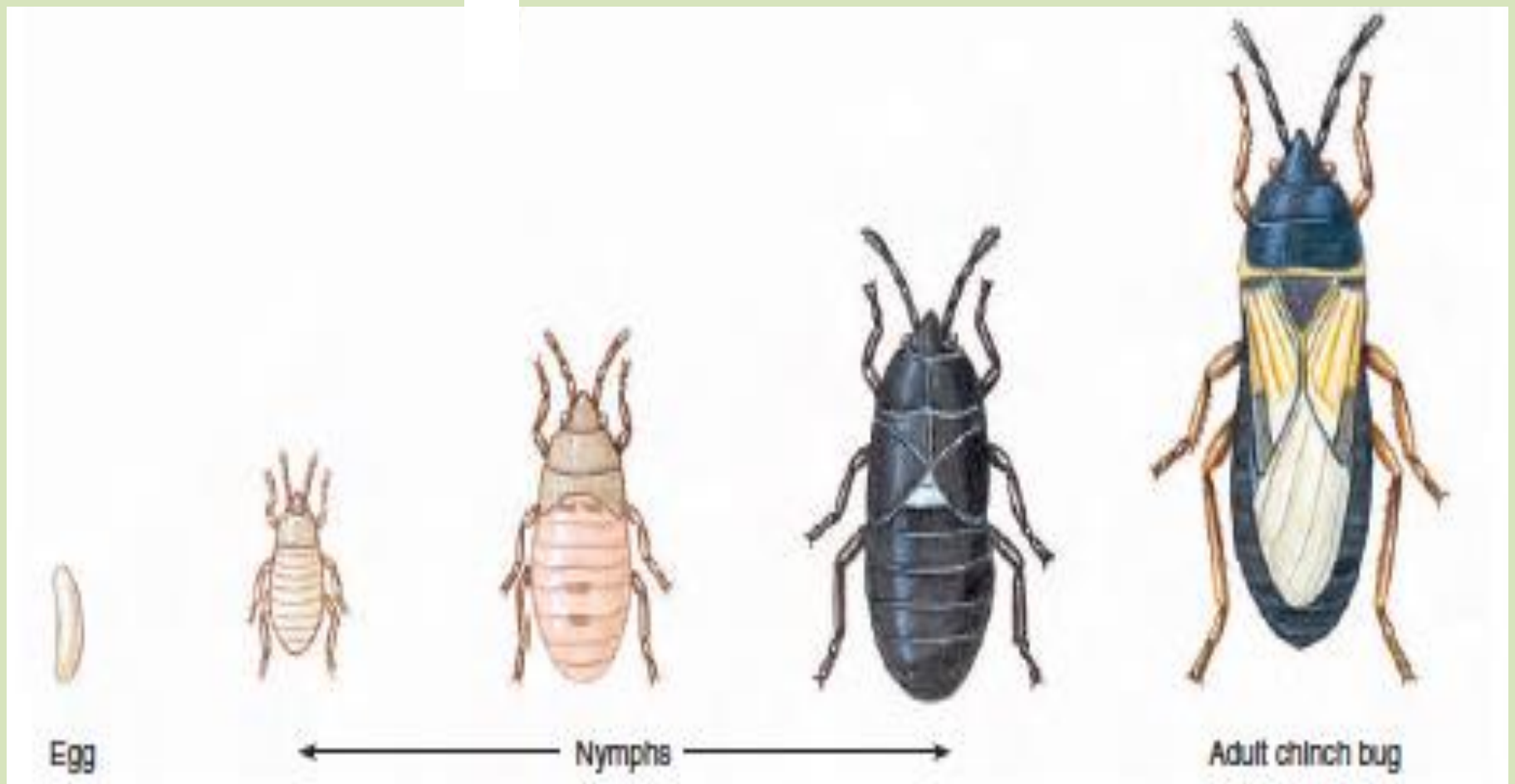
SUBPHYLUM: HEXAPODA, CLASS: INSECTA

Holometabolous metamorphosis

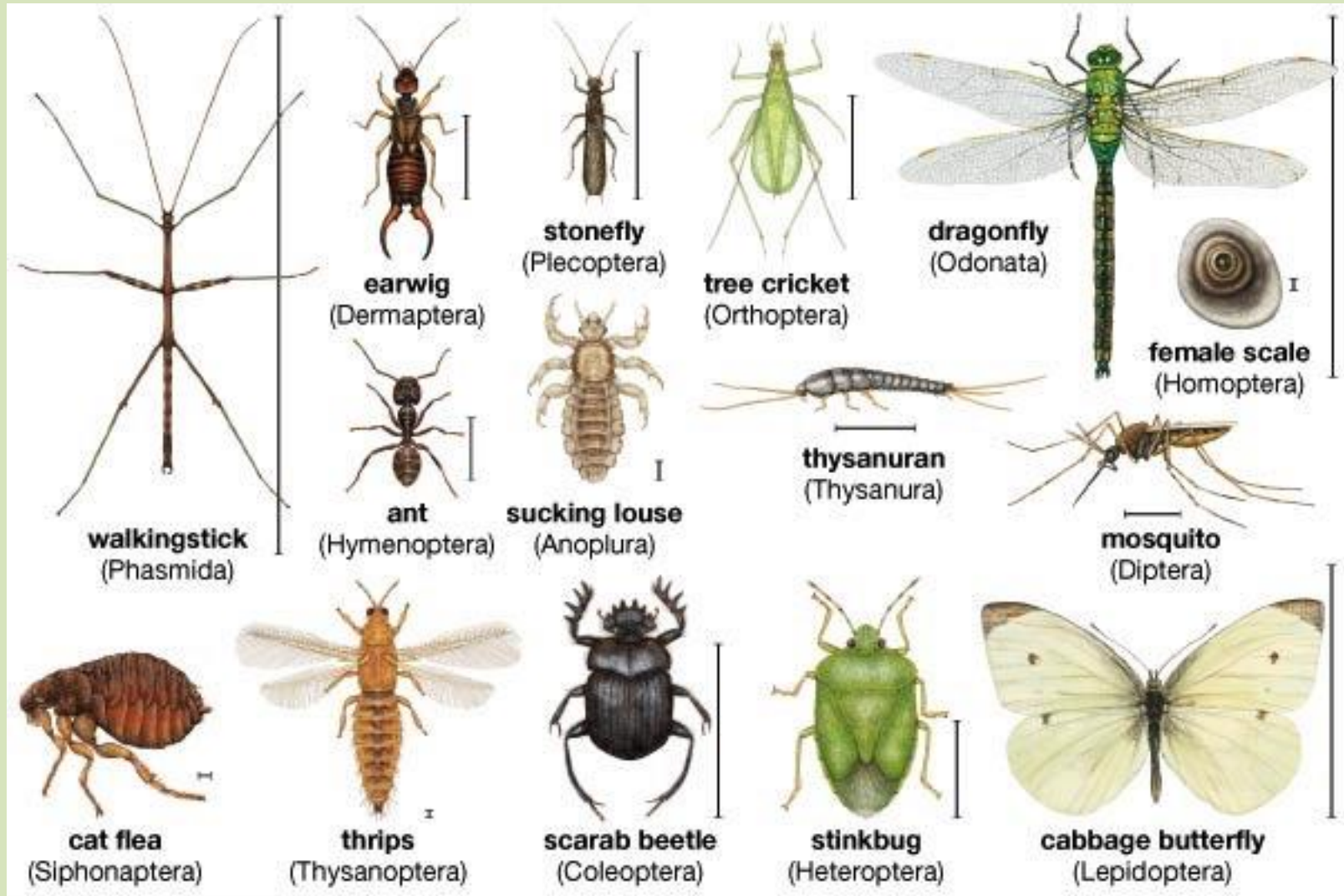


SUBPHYLUM: HEXAPODA, CLASS: INSECTA

Hemimetabolous metamorphosis



SUBPHYLUM: HEXAPODA, CLASS: INSECTA



THANK YOU!

